# STAFF REVIEW · H446 018 source models

Separate from the eight-page pupil workbook. This copy is in a public repository and is NOT access controlled. Check technical correctness before use.

Printed question mapping: Booklet Q1 = original online Q1; Booklet Q2 = original online Q4

## Do Now — source answers

1. The third element. Array indexes start at 0.

2. The size of dimension 1 — the number of columns.

3. An array.

4. The data item has several related fields that belong together.

## I Do — source worked steps

1. Declare the record type — CODE:
public record Student(string Name, int CandidateNumber, int Score);

2. Create a student — CODE:
Student pupil = new Student("Amelia", 1037, 72);

3. Access a property — CODE:
Console.WriteLine(pupil.Score); // 72

4. Update using with — CODE:
Student updated = pupil with { Score = 80 };
Console.WriteLine(updated.Score); // 80

## We Do — guided CHECK

1. Fill the same code
CODE:
names.Add("Daisy");
names.Remove("Ben");

2. Fill the same code
CODE:
var result = (Name: "Amelia", Score: 72);
Console.WriteLine(result.?);
Console.WriteLine(result.?);
CODE:
var result = (Name: "Amelia", Score: 72);
Console.WriteLine(result.Name);
Console.WriteLine(result.Score);

3. Complete the decision
CODE:
Need ordered items
Need add/remove
Size changes

Best structure = ?
CODE:
Need ordered items
Need add/remove
Size changes
Best structure = List

4. Complete the fields
CODE:
Book record
Title = ?
Author = ?
ISBN = ?
CODE:
Book record
Title = "Dune"
Author = "Frank Herbert"
ISBN = "978..."

### You Do — source mark schemes

1. Create and edit a list
CODE:
```
using System;
using System.Collections.Generic;

public class Program
{
public static void Main()
{
List<string> names = new List<string> { "Amelia", "Ben", "Charlie", "Daisy" };
names.Add("Ethan");
names.Remove("Ben");

foreach (string name in names)
{
Console.WriteLine(name);
}
}
}
```
Check: The list changes size and every remaining item is displayed.

2. Group a result
CODE:
```
using System;

public class Program
{
public static void Main()
```

```
{
var result = (Name: "Amelia", Score: 72);
Console.WriteLine(result.Name);
Console.WriteLine(result.Score);
}
}
```
Check: Both tuple fields are accessed by name.

3. Find an item
CODE:
```
using System;
using System.Collections.Generic;

public class Program
{
public static void Main()
{
List<string> courses = new List<string> { "Maths", "Physics", "Computer Science", "History", "Art"
};
Console.Write("Course: ");
string wanted = Console.ReadLine();

if (courses.Contains(wanted))
Console.WriteLine("Found");
else
Console.WriteLine("Not found");
}
}
```
Check: A present and absent course both produce sensible output.

4. Explain the structure
CODE:
```
// Array: fixed-size ordered collection.
// List: ordered collection that can grow or shrink.
// Record: a reusable named type with related properties.
// Tuple: a small fixed group of related values.
```
Check: Each choice refers to the behaviour of the structure, not just its name.

## Exit Ticket — source answer

1. True. A C# List<T> can add and remove items as the program runs.